



Geographical Analysis in SAP Business Information Warehouse

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ABSTRACT

The visualization of business data in a geographic context supports the decision making process by showing how e.g. sales data relates to other data. Tasks like *market penetration analysis* can be "visually" tackled by combining enterprise and market data on a map. In this paper it is shown how maps can be linked to the data queried from SAP's Business Information Warehouse (BW), in order to benefit from geographic analysis. BW provides a seamless integration of its unique GIS Tool called *Business Explorer Map* (BEx Map), in such a way that any report containing geographic data, can be displayed on a map. BEx Map provides an easy way of navigating (zoom in, geographic drill-down, slice & dice) through BW reports, enabling all end-users to interactively analyze the BW report data.

Keywords

Geographic Interface, Data Warehouse, OLAP, Visualization.

1. INTRODUCTION

Decisions today are based on a growing number of figures and facts. The visualization of these business figures is getting more and more important, as it helps to quickly scan and understand the data. Geographic Information System (GIS) technology integrates business reporting with the unique visualization and geographic analysis benefits offered by maps, such that relationships in the data become more apparent. However, the business data residing in a data warehouse has to be tightly linked to geographical maps, in order to enable interactive analysis in a GIS.

The main contribution of this paper is, to show how SAP Business Information Warehouse (BW) and GIS are combined, and what benefits result from such an integration. In section 2 the central role of SAP's data warehouse solution is discussed and the reporting tool is presented. Section 3 is concerned with GIS tools specialized for business applications. Section 4 introduces the *Business Explorer Map* - a unique GIS tool for multidimensional

database queries. Finally, in section 5 the business impact of the *Business Explorer Map* is concluded.

2. SAP Business Intelligence

Although the Internet is redefining the way business is done today, one thing remains crucial to the success of a company: the value of interpreted information. However, getting hold of important facts and figures can be a complex and time-consuming task, as the information is spread throughout the corporate IT structures on a variety of platforms and applications.

The *mySAP.com*TM [5] concept allows to implement an integrated view across the entire e-business intelligence architecture including systems, data and process flows. And, importantly, enabling closed-loop scenarios. *mySAP.com* integrates a large number of business applications (e.g. Financial Accounting, Controlling, Sales and Distribution, Materials Management, Production Planning, etc.) that share data and basic business objects in a central database [7]. SAP Business Intelligence (BI), which is included in *mySAP.com*, enables real-time analysis and decision making in e-business environments. It comprises a data warehouse (SAP BW) component as the platform for eBusiness Intelligence plus packaged, e-ready and ERP-integrated analytical applications (Enterprise Intelligence, Customer Relation Intelligence, Supply Chain Intelligence).

The SAP Business Information Warehouse [2] provides analytic application packages that offer a complete business solution for analyzing and reporting on information in specific business areas. They measure business operations' effectiveness and help you make tactical and strategic business decisions intended to improve profits, reduce costs, and increase competitiveness. SAP BW provides interfaces that enable it as a platform for e-business intelligence, independently of the environment - SAP, mixed or non-SAP. SAP BW consists of a preconfigured meta data repository, a warehouse management tool, automated data extraction and staging tools, a relational OLAP (Online Analytical Processing) engine [3], and a Front End with powerful reporting and analysis capabilities (Business Explorer) [2].

SAP BW allows the end-user to look at the data from many perspectives and combine data from various sources. One can drill-down from one level of detail to another. The OLAP processor guarantees that all business rules are satisfied. BW Reports are interactive, in such a way that one can move from one general view to one which is more detailed (slicing or drill-down), or from a very detailed level to one which is more aggregated. One can also decide to view data from a different perspective (dicing).

Performance is of crucial importance for any data warehouse to be accepted by the end-users. The architecture of SAP BW

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includes several features to ensure high performance. This also results from the fact that it is separated from the OLTP (Online Transaction Processing) system [2].

3. GIS in Business Applications

"Better information leads to better decisions." This is as true for GIS as it is for other information systems. A GIS, however, is not an automated decision making system, but a tool to query, analyze, and map data in support of the decision making process. The visualization of business data helps to better understand the connection in the data. Displaying business data in a geographical context, i.e. on a geographic map, clearly shows the neighbourhood relationships, and thus contributes to the decision making process. Tasks like *market penetration analysis* or the search for *new markets* can be "visually" tackled by combining enterprise and market data.

Standard Geographic Information Systems, however, are historically designed for statistical and scientific tasks. One needs special training, in order to effectively use a GIS with complex functionality. GIS projects usually need to be configured by GIS experts (link database, map layers etc.). These tasks cannot be accomplished by business end-users, who are interested in ad hoc reports with dynamic query views. In addition, standard GIS diagrams do not satisfy business charting needs. It should also be noted, that cartography has not yet fully recognized the demand for pure business charting.

To the author's knowledge, there are only a few GIS tools (e.g. ESRI's Business Analyst [1], Macon's Regiograph 5.0 [6]), that really support business visualization and provide more sophisticated graphic renderers than simple bar and pie charts. Business users, however, are typically interested in performance indicator graphs like gauge controls (speedometer), split pies, stacked bars, and portfolio diagrams. In addition, these GIS tools do not provide ready-to-use interfaces for generic OLAP queries.

4. SAP BW Business Explorer Map

For interactive ad hoc data exploration in a geographic context, it is necessary to tightly link maps to the data. Standard GIS tools support persistent links to attribute tables (e.g. on a local PC drive). However, for multidimensional data residing in a data warehouse a specific interface for exploring the data is needed. The "Geo-Query" would ideally be formulated on the map itself, by intuitively selecting geographical regions and corresponding slices from the data cube.

Therefore, SAP BW's Business Explorer Map (BEx Map) is carefully integrated into both the BW data model [2] and the BW Front-End. In the BW server, one has the option to mark individual dimensions (InfoObject) of the BW Cube as "geo-relevant" (see Figure 1). Typical candidates for such geo-dimensions are e.g. *Country*, *Region*, or *Zip Code*. For each geo-dimension it is necessary to upload maps, that are in the ESRI Shapefile format [4]. In order to link BW report data and map shapes, the Shapefiles need to be extended with BW master data keys for the given geo-dimension. The Shapefiles are transferred from the BW Server to the Front-End client on demand having the benefit of the central management of (up-to-date) maps. However, for point data like e.g. *Customer*, *Sales Office*, or *Store location* the use of static Shapefiles is not desirable. (If a customer moves or new ones are added, a new Shapefile must be created and uploaded.) In order to display the most recent location data, BEx Map gets

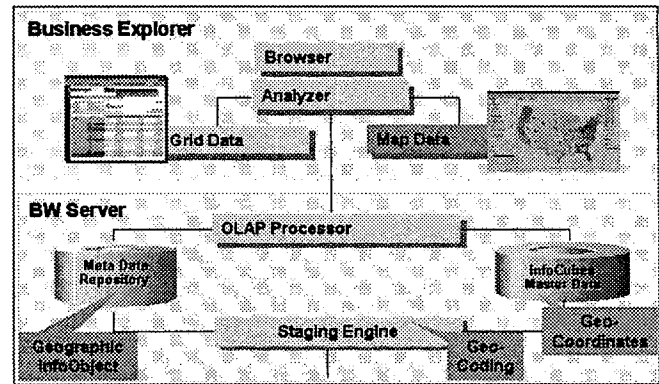


Figure 1. Integration of GIS in the Business Information Warehouse

the geocoded coordinates (see Figure 1) of e.g. customers directly from the BW master data.

Much effort was put in the interface design, in order to present an easy to use and intuitive geographic analysis tool. The central part of the BEx Map interface (Figure 2) contains the map layers. In the Map Design section (on the left), the "Geo-Query" for each layer is defined. The legend on the right side dynamically displays the value ranges queried from the BW server.

End-users do not necessarily need to understand GIS technology or care for details like loading map layers and manually link attribute data to the layers. This is taken care of by BEx Map: any BW report containing geographic data is analyzed, in order to identify geographic dimensions and automatically load the corresponding maps from the server.

After a Map is attached to a BW report, a "Geo-Query" can easily be formulated using a natural language metaphor, e.g.: "For the *country* level: Show me *Sales* by *Month* using *Bar Charts*, over (color-shaded) *Market Potential*."

Analogous to the drill-down feature (slice & dice) in the BW report, it is also possible to do geographical drill-downs from one map layer to a more detailed layer. For example a double click on California in Figure 2, would drill-down to the next layer, resulting in the same graphs (sales as bar chart over color-shaded market potential) for the CA Regions.

In addition, point data like customer locations and sales offices can be visualized on the fly: the most recent coordinates are extracted from geo-coded BW Master Data. One can quickly find out the regions where the customers are located. Standard GIS functionality like: Zoom in/out, Pan, Full Extent, and setting Layer Properties is available via the GIS toolbar on top of the map. The standard layer technique is implemented: layers are ordered by level of detail, in such a way that the more detailed layers are on top. The user can switch on/off layers by clicking on the buttons beneath the map. Several graphic renderers are available for each map layer: Color-Shading, Pie/Bar charts, and Symbols. Other business specific renderers like Split Pies, Stacked Bars, Portfolio diagrams, and Dot Density will be integrated. Finally, after the user is satisfied with the geographic analysis, and maybe has identified regional problems, she can directly jump to the tabular view of the report and investigate further by slicing and dicing on non-geographical dimensions.

The real value of business GIS, however, lies in the combination of enterprise data and market data (AC Nielsen, Dun & Brad-

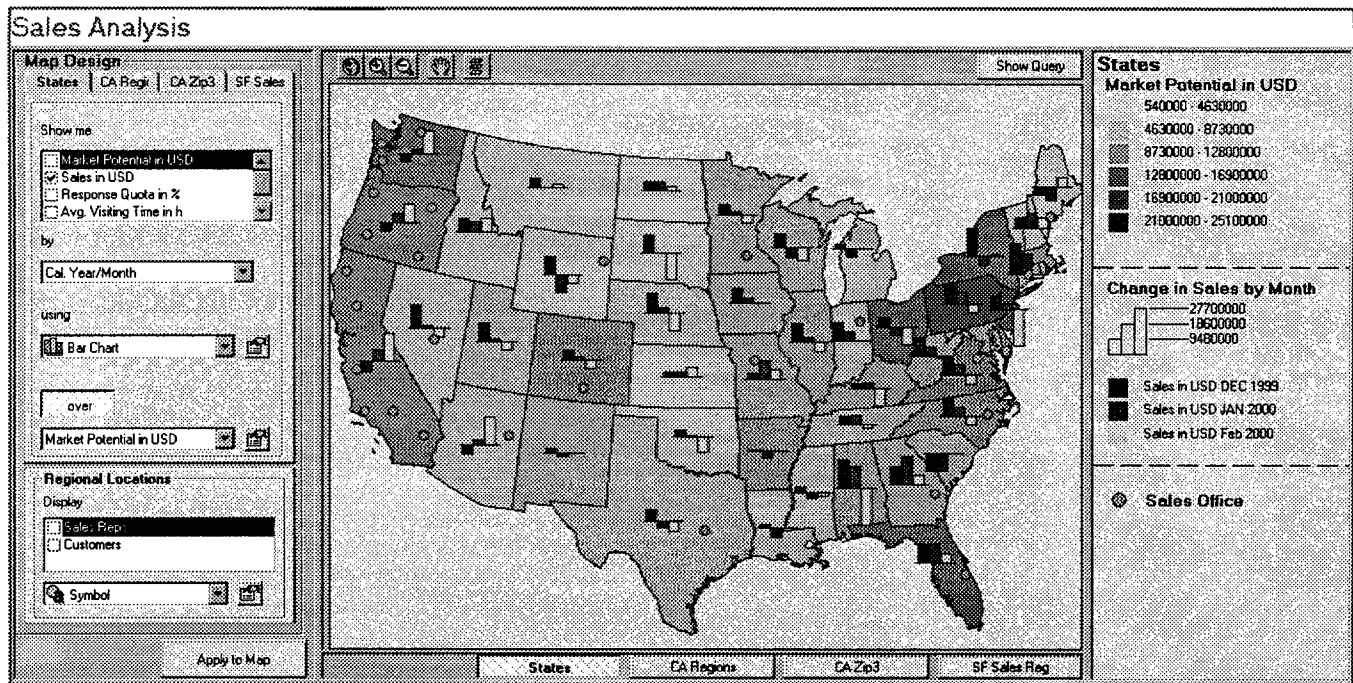


Figure 2. Business Explorer Map User Interface.

street, demographic data). Market data can easily be added to enterprise data in BW and thus visualized in BEx Map.

The ultimate success of a data warehouse is reflected by the number of end-users who benefit from its offerings. BEx Map enables all end-users to explore BW reports, and thus also contributes to the success of BW.

5. Conclusion

The visualization of business data on geographic maps, helps to better understand the connection in the data and thus supports the decision making process. The seamless integration of geographic maps into SAP Business Information Warehouse reports presented here, allows for flexible and generic use of Business GIS analysis. We believe that the intuitive and easy to use interface of BW Business Explorer Map enables all end-users to interactively explore multidimensional BW reports in a geographic context.

6. REFERENCES

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